Aparajita Gupta

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Education

Purdue University **Graduating May 2028**
Bachelor of Science in Computer Engineering, Concentration in AI/ML & Computer Systems *West Lafayette, IN*

- Relevant Coursework: Electrical Engineering Fundamentals, Microprocessor Systems and Interfacing, Digital System Design, Signals and Systems, Data Structures, Python for Data Science, Advanced C Programming, Linear Algebra, Multivariate Calculus, Ordinary Differential Equations, Discrete Math
- Activities/Societies: IEEE Motorsports, Data Mine, Embedded Systems @ Purdue, Society of Women Engineers

Experience

IEEE Motorsports **Jan 2025 – Present**
Electrical Engineer *Purdue University*

- Contributed to electrical design of a 24s8p, 192-cell battery system with 36 backup cells for a student-built EV race car
- Minimized power loss to 5W by integrating nickel strips and BMS with thermal management for safe motor control
- Redesigned automated precharge circuit to operate with 0 chatter or lag, replacing manual switch-based charging
- Earned 1st Place in Battery Efficiency at the EV Grand Prix with a 4,320 Wh, 86.4V battery sustaining a 55-min race

Inari - Digital Phenotyping with Soybean Imagery **Aug 2025 – May 2026**
AI/ML Developer (ML + Data Science) *Purdue University*

- Linked plant phenotypes to genotypes, achieving accuracy of 96.4% and an F1 score of 0.965 using ResNet-18 CNN
- Extracted key physical plant traits by developing phenotyping algorithms to process and analyze 8K soybean images
- Optimized dataset quality by conducting EDA, OTSU/SAM segmentation, and feature extraction (DGCI, MPRI, HSV)
- Identified top predictor by evaluating 5+ ML models in Scikit-learn & TensorFlow (ResNet-18 CNN, KNN, SVM, RF)

The Data Mine **Aug 2024 – May 2026**
Undergraduate Data Science Researcher *Purdue University*

- Analyzed 10+ real-world datasets using statistical techniques and visualizations to generate actionable insights
- Performed full-stack data analysis with R, Python, SQL, Bash, Polars, PyTorch, and MongoDB for cleaning/modeling
- Collaborated in a 12-member Agile team, improving deliverable quality & team communication through biweekly sprints
- Delivered technical reports with self-assessments and future goals, delivering findings implemented by industry partners

Projects

3-Band Analog Audio Equalizer | *LM356, LM386, RC Filters, Oscilloscope* **Analog & Signal Processing**

- Designed & built a 5-stage analog equalizer with bass, mid, and treble controls via passive RC filters and op-amps
- Verified the 3 filter cutoff frequencies all well under ±10% tolerance via oscilloscope frequency response analysis
- Achieved < 1.26 mVRMS output ripple across full 100 Hz–10 kHz spectrum, confirming flat and consistent equalization
- Drove an 8Ω speaker at 578–601 mW output via LM386 power amplifier across all tested frequencies

BrightGrade | *Python, Playwright, Google Gemini API, Todoist REST API, AppleScript* **Backend Automation**

- Automated deadline tracking through a dual-source pipeline for 300+ tasks across 7 courses, resolving 88% of due dates
- Integrated Gemini API to auto-generate task titles and parse syllabus PDFs, matching user style across 200+ tasks
- Fixed bugs by validating against live account data, correcting 35+ real tasks & reducing false-positive noise by 50%
- Delivered live results via Todoist & AppleScript-driven Calendar.app events, syncing exams & tasks

SprinkNet | *IoT, MicroPython, Swift, ESP32, MQTT* **Embedded Systems**

- Automated a 4-zone home sprinkler system with ESP32 MicroPython firmware that switches a GPIO relay for each zone
- Scheduled 4 independent zone cycles alongside live app commands using asyncio tasks and NTP-synced cron timing
- Enabled remote control of every zone in < 1s via a Swift iOS app that pushes commands and schedules over MQTT
- Ran 12+ months with 0 physical reflashes by adding over-the-air updates from GitHub and MQTT auto-reconnect

Technical Skills

Languages: C, C#, Python, Java, Bash, MATLAB, System Verilog
Libraries/Frameworks: NumPy, Pandas, Matplotlib, MongoDB, GitHub, React.js, Keras, Linux/Unix shell scripting
Engineering Skills: Circuit design, breadboarding, soldering, LTspice, KiCAD, FPGA & IoT programming, Excel